

FORM TP 2013023



TEST CODE **01238032**

JANUARY 2013

CARIBBEAN EXAMINATIONS COUNCIL

**CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION**

PHYSICS

Paper 032 – General Proficiency

Alternative to SBA

2 hours 10 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. You **MUST** use this answer booklet when responding to the questions. For each question, write your answer in the space provided and return the answer booklet at the end of the examination.
2. **ALL WORKING MUST BE SHOWN** in this booklet, since marks will be awarded for correct steps in calculations.
3. Attempt **ALL** questions.
4. You may use a silent, non-programmable calculator.
5. Mathematical tables are provided.
6. You are advised to take some time to read through the paper and plan your answers.

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DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

Attempt ALL questions.

You MUST write your answers in this answer booklet.

1. A motor cyclist travels from St John's to the Vivian Richards Stadium in Antigua. His velocity was recorded at intervals for 60 s as shown in Table 1.

TABLE 1

Velocity, $v/\text{m s}^{-1}$	0	8	14	29	30	37	45
Time, t/s	0	10	20	30	40	50	60

- (a) On page 3, plot a graph of velocity versus time. (11 marks)

- (b) One of the readings was incorrectly recorded. Identify this reading and write a possible value.

(2 marks)

- (c) Calculate the gradient of the graph.

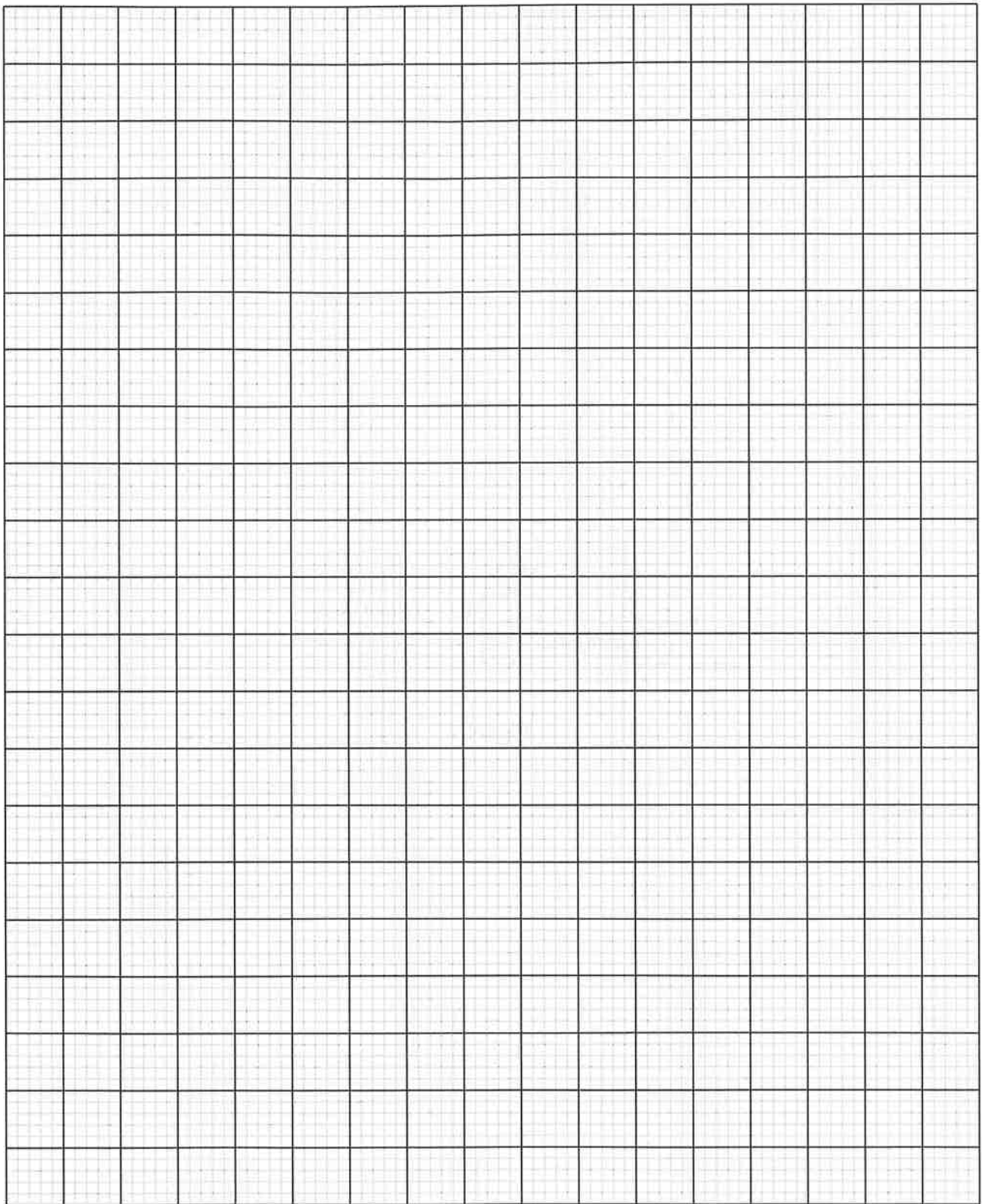
(6 marks)

- (d) What physical quantity does the gradient represent?

(1 mark)

Total 20 marks

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2. (a) Boyle's law states that the volume of a fixed mass of gas is inversely proportional to its pressure if the temperature is constant.

Students in a fifth form class were given the apparatus shown in Figure 1 and told to use it to investigate this law.

Label Figure 1 and describe how the investigation may be done.

Include ONE precaution that should be taken to keep the temperature constant.

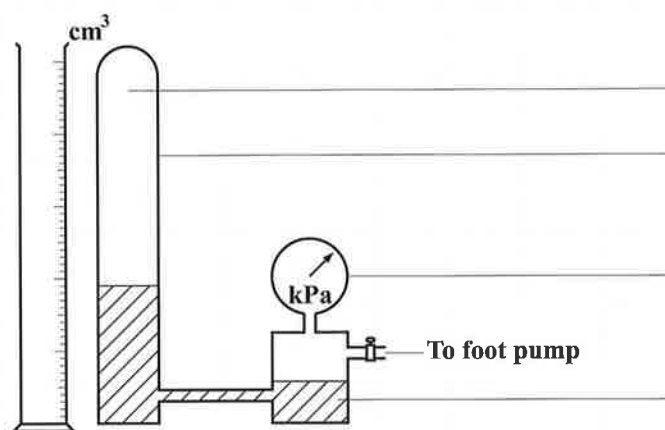


Figure 1

(10 marks)

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- (b) The students' results were plotted on the graph shown in Figure 2.

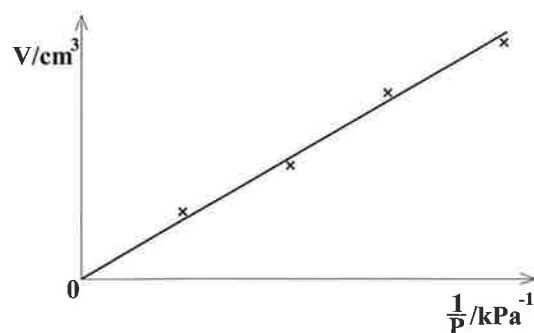
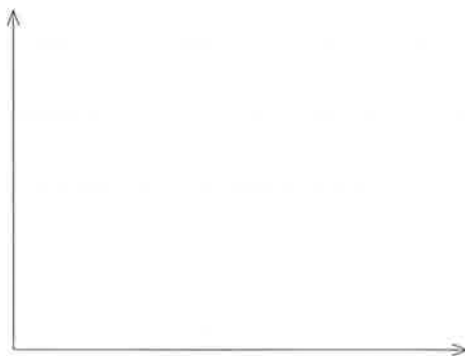


Figure 2

Explain whether the graph shows that Boyle's law is confirmed or not.

(4 marks)

- (c) If the students had plotted instead a graph of V against P , **sketch** the graph they would have obtained.



(2 marks)

Total 16 marks

3. (a) Jamal and Kim were asked to demonstrate, using a magnet with unmarked poles, how they could distinguish the North Pole from the South Pole. Describe, with the aid of a diagram, a method they might have used.

Diagram(s):

(2 marks)

Procedure:

(4 marks)

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- (b) Jamal and Kim were also given two identical unmarked steel bars and told that one of them was a magnet while the other was a magnetic material. Show how they could identify which bar was the magnet.

(6 marks)

Total 12 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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